1. Steps for Execution of Project

Follow each step in the exact order listed below. Two terminal windows will be required — one for the backend and one for the frontend.

Step 1 — Copy Project from CD / USB

1. Insert the CD/USB into your computer.
1. Navigate to the CD/USB drive using File Explorer (Windows) or Finder (macOS).
1. Copy the entire FluxAi project folder to a convenient location on your computer (e.g., Desktop or Documents).

Step 2 — Extract the Project Files

1. Locate the copied project folder or ZIP file.
1. If it is a ZIP archive, right-click it and select Extract All (Windows) or double-click to extract (macOS).
1. Ensure the extracted folder is named FluxAi/ and is accessible.

Step 3 — Open the Project in VS Code

1. Open Visual Studio Code.
1. Click File → Open Folder and select the extracted FluxAi folder.
1. The project structure should be visible in the left sidebar Explorer panel.

Step 4 — Backend Setup & Configuration

Open a terminal inside VS Code (Terminal → New Terminal) or use your system terminal.

1. Navigate to the backend folder:
```
cd chatbot_backend-main/backend
```
1. Install all backend dependencies:
```
npm install
```

Wait for the process to complete. A node_modules folder will be created.

1. Create the environment configuration file. In the backend folder, create a new file named exactly:
```
.env
```
1. Open the .env file and add the following configuration (replace placeholder values with your actual credentials):

Key	Value / Example	Description
PORT	5000	Backend server port
MONGODB_URI	mongodb+srv://...	Your MongoDB Atlas connection string
JWT_SECRET	any_random_string	Secret key for authentication tokens
LLM_PROVIDER	groq (or google/openai)	AI engine to power the chatbot
GROQ_API_KEY	Obtain from groq.com	API Key for Groq LLM
CLOUDINARY_CLOUD_NAME	Your cloud name	For document file hosting
CLOUDINARY_API_KEY	Your API key	Cloudinary API Key
CLOUDINARY_API_SECRET	Your API secret	Cloudinary API Secret
LLAMAPARSE_API_KEY	From llamaparse website	For PDF/DOCX text extraction

1. After configuring .env, seed the default admin user by running:

```
npm run seed
```

This creates the default administrator account in the database.

Step 5 — Frontend Setup

Open a second terminal window (keep the first one open).

1. Navigate to the frontend folder:

```
cd chatbot_frontend-main/frontend
```

1. Install all frontend dependencies:

```
npm install
```

Wait for the process to complete. This may take a few minutes.

Step 6 — Start the Backend Server

In the first terminal (inside the backend folder), run:

```
npm run dev
```

If successful, the terminal will display: Server running on port 5000 and Connected to MongoDB. Keep this terminal open and running.

Step 7 — Start the Frontend Application

In the second terminal (inside the frontend folder), run:

```
npm run dev
```

The terminal will display the local URL. Keep this terminal open.

Step 8 — Open the Application in Browser

1. Open Google Chrome (recommended) or any modern web browser.
1. In the address bar, enter:

```
http://localhost:5173
```

1. The FluxAI login page will appear. Use the following default Admin credentials to log in:

Field	Value
Email	admin@example.com
Password	admin123
Role	Administrator

4. Troubleshooting Guide

If you encounter any issues during setup or execution, refer to the quick-fix table below.

Problem / Error	Solution
MongoDB not connecting	Verify the MONGODB_URI in .env is correct. Ensure your IP is whitelisted in MongoDB Atlas (Network Access → Add IP Address: 0.0.0.0/0).
Module not found error	Delete the node_modules folder inside the affected directory and run npm install again.
Port 5000 or 5173 already in use	Change the PORT value in the .env file for the backend or use a different port for the frontend.
Missing API Key in AI Chat	Ensure LLM_PROVIDER is set and the matching API key (e.g., GROQ_API_KEY) is filled correctly in .env.
Login page not appearing	Confirm both the backend (port 5000) and frontend (port 5173) servers are running simultaneously in separate terminals.
npm install fails / hangs	Check your internet connection. Then delete node_modules and package-lock.json, and run npm install again.

5. Important Notes

Internet Required: An active internet connection is mandatory — for MongoDB Atlas database access and AI API calls.

API Keys Required: Obtain the following API keys and add them to your .env file before starting the project.

- Groq API Key — obtain from: https://groq.com (free tier available)
- Cloudinary API Key, Cloud Name & Secret — from: https://cloudinary.com (free account)
- LlamaParse API Key — for document text extraction (PDF/DOCX parsing)
- OpenAI or Gemini API Key — only required if you choose openai or google as LLM_PROVIDER

Security Notice: Never share your .env file or upload it to any public platform (e.g., GitHub). It contains sensitive API keys and credentials.

Component	Default Port
Backend Server	http://localhost:5000
Frontend Application	http://localhost:5173